**1. Network Reconnaissance using Nmap Identify open ports, running services, and OS fingerprinting of target machine in a lab network. Tool: Nmap**

**1. Install Nmap**

sudo apt install nmap -y

**2. Check your IP address**

ip a

**3. Discover devices in the network**

sudo nmap -sn 192.168.1.0/24

**4. Scan open ports**

nmap -p- TARGET_IP

**5. Identify running services and versions**

nmap -sV TARGET_IP

**6. Detect operating system**

sudo nmap -O TARGET_IP

**7. Complete scan**

sudo nmap -A TARGET_IP

**2. Packet Analysis using Wireshark Inspect TCP handshake, HTTP requests, DNS queries and detect suspicious patterns. Tool: Wireshark**

**STEP 1: Install Wireshark**

Open Kali Terminal:

sudo apt install wireshark -y

**STEP 2: Open Wireshark**

sudo wireshark -i eth0

1. Wireshark will open.
2. Select **eth0** (your network interface).
3. Double-click eth0 to start capturing packets.

**STEP 3: Generate Network Traffic**

Open another Kali terminal and execute:

**1. Generate DNS query**

nslookup google.com

**2. Generate HTTP request**

curl http://neverssl.com

**3. Generate TCP traffic**

curl https://google.com

**STEP 4: Apply Wireshark Filters**

Enter these filters in the top filter bar one by one.

**1. TCP packets**

tcp

**2. TCP Three-way Handshake**

tcp.flags.syn == 1

Observe:

- SYN
- SYN, ACK
- ACK

**3. HTTP requests**

http

**4. DNS queries**

dns

**5. Suspicious TCP connections**

tcp.flags.reset == 1

This displays TCP reset packets.

**STEP 5: Stop Capturing**

Click the red square **Stop** button in Wireshark.

3. **Firewall Configuration in Linux (iptables / ufw) Create rules to block specific ports and restrict SSH access.**

**STEP 1: Install UFW**
sudo apt install ufw -y
**STEP 2: Check firewall status**
sudo ufw status
**STEP 3: Enable firewall**
sudo ufw enable
**STEP 4: Block specific ports**
Block HTTP (Port 80):
sudo ufw deny 80/tcp
Block FTP (Port 21):
sudo ufw deny 21/tcp
**STEP 5: Allow SSH access**
Allow SSH only from your Windows PC:
sudo ufw allow from 192.168.1.43 to any port 22 proto tcp
Block SSH from other devices:
sudo ufw deny 22/tcp
**Important:** Add the specific allow rule before the deny rule. If you are connected remotely through SSH, don't enable these rules until you have verified your access.
**STEP 6: View firewall rules**
sudo ufw status numbered
**STEP 7: Delete a firewall rule (if required)**
sudo ufw delete deny 80/tcp
**STEP 8: Disable firewall (optional)**
sudo ufw disable

4. **Sniffing and ARP Spoofing Demonstration Capture traffic between two machines using ARP poisoning. Tools: Ettercap, arpspoof**

**STEP 1: Install Ettercap**

Open Kali Terminal:

sudo apt update

sudo apt install ettercap-graphical -y

**STEP 2: Find Network IPs**

ip a

Check gateway IP:

ip route

Note your Kali IP, Windows IP and gateway IP.

**STEP 3: Open Ettercap**

sudo ettercap -G

**STEP 4: Start Network Sniffing**

1. Click **Sniff → Unified Sniffing**.
2. Select eth0.
3. Click OK.
4. Go to **Hosts → Scan for hosts**.
5. Click **Hosts → Hosts List**.

**STEP 5: Select Lab Devices**

For an authorized isolated lab:

1. Select your first test machine.
2. Click **Add to Target 1**.
3. Select your second test machine.
4. Click **Add to Target 2**.

**STEP 6: Demonstrate ARP Poisoning**

1. Open **Mitm** menu.

2. Select **ARP Poisoning**.
3. Enable **Sniff remote connections**.
4. Start the demonstration.

Only do this between devices you own in an isolated lab. ARP poisoning can interrupt connectivity and expose unencrypted traffic.

**STEP 7: Observe Traffic**

Generate harmless test traffic from your lab machines.

Observe captured packets in Ettercap.

You can also open Wireshark and use this display filter:

arp

To view TCP traffic:

tcp

**STEP 8: Stop the Demonstration**

1. Stop Ettercap sniffing.
2. Close Ettercap.
3. Check that both test devices have normal network connectivity.

**Result**

Successfully demonstrated ARP poisoning and network traffic sniffing using Ettercap in an authorized lab environment.

**5. SQL Injection Attack Demonstration**

**Bypass login authentication in a vulnerable test application. Platform: DVWA**

**(Damn Vulnerable Web App)**

**STEP 1: Install Required Packages**

Open Kali Terminal:

sudo apt update

sudo apt install apache2 mariadb-server php php-mysqli php-gd php-curl php-xml git -y

**STEP 2: Start Services**

sudo service apache2 start

sudo service mariadb start

**STEP 3: Download DVWA**

cd /var/www/html

sudo git clone https://github.com/digininja/DVWA.git dvwa

Set permissions:

sudo chmod -R 755 /var/www/html/dvwa

**STEP 4: Configure DVWA**

cd /var/www/html/dvwa/config

sudo cp config.inc.php.dist config.inc.php

Create database and user:

sudo mariadb -e "CREATE DATABASE dvwa; CREATE USER 'dvwa'@'localhost' IDENTIFIED BY 'p@ssw0rd'; GRANT ALL PRIVILEGES ON dvwa.* TO 'dvwa'@'localhost'; FLUSH PRIVILEGES;"

**STEP 5: Open DVWA**

Open Firefox in Kali and visit:

http://127.0.0.1/dvwa/setup.php

1. Find **Create / Reset Database**.
2. Click the button.

3. Wait for database setup.

**STEP 6: Login to DVWA**

Open:

http://127.0.0.1/dvwa/login.php

Default credentials:

Username: admin

Password: password

**STEP 7: Set Security Level**

1. Click DVWA Security.
2. Select **Low**.
3. Click Submit.

**STEP 8: Perform SQL Injection**

1. Click **SQL Injection** in the left menu.
2. Enter this value in the User ID field:

1' OR '1'='1' #

3. Click Submit.

**Observation:** The application may display multiple user records instead of only one because the injected condition changes the SQL query logic.

**STEP 9: Understand the Attack**

Normal SQL query:

SELECT * FROM users WHERE user_id = '1';

Injected condition:

' OR '1'='1'

**Practical: Intrusion Detection using Snort**

**Aim:** To detect suspicious network traffic using custom Snort rules.

**STEP 1: Check Snort installation**

**Sudo apt install snort**

snort -V

**STEP 2: Create custom rule**

sudo mkdir -p /etc/snort/rules

Open the rule file:

sudo nano /etc/snort/rules/local.rules

Add this rule:

alert icmp any any -> any any (msg:"ICMP Traffic Detected"; sid:1000001; rev:1;)

Save using Ctrl + O, Enter, Ctrl + X.

**STEP 3: Find Snort configuration**

sudo find / -name snort.lua 2>/dev/null

Normally, it should show:

/etc/snort/snort.lua

**STEP 4: Run Snort**

If the configuration exists at /etc/snort/snort.lua, execute:

sudo snort -c /etc/snort/snort.lua -R /etc/snort/rules/local.rules -i eth0 -A alert_fast

**STEP 5: Generate network traffic**

Open another Kali terminal:

ping -c 4 192.168.1.43

**STEP 6: Observe alerts**

Return to the Snort terminal.

Expected alert:

ICMP Traffic Detected

Press Ctrl + C to stop Snort.

**STEP 7: Check Snort logs**

sudo ls /var/log/snort/

---

**Result**

Successfully installed and executed Snort 3, created a custom ICMP detection rule and monitored network traffic for suspicious activity.

## Cross-Site Scripting (XSS) Attack Demonstration

## index.html:

```
<html>

<body>

<h2>XSS
Demonstration</h
2> Enter Name:
<input id="name">

<button onclick="show()">Submit</button>

<p id="output"></p>

<script>

function show() {

                                        var name =
    document.getElementById("name").value;

  document.getElementById("output").innerHTML = name;

}

</script>

</body>
```

**Create Website:**

```
mkdir xss_testing
cd xss_testing
nano index.html
nano index.html
```

**Test Xss:**

**Inside Enter name txtbox:**

```
<img src=x onerror="alert('You are hacked')">
```

TO prevent this XSS replace:

.innerHTML=name

With

.textContent = name;

**DNS Enumeration and WHOIS Lookup**

Step 1
search **who.is**
Step 2
Search any website(ex. Amazon.com)
Step 3
Scroll down to see Contact Information & Name Servers
Step 4

Select option “DNS records for DNS information”

**Vulnerability Scanning using OpenVAS/Nessus(Community Edition)**

Step1:

Command:

**Ifconfig**

Step2:

Command:

**Ping 'ip' (eth0)**

Step3:

Command:

**Nmap -sV 'ip' (eth0)**

Step4:

Command:

**Msfconsole**

Step5:

Command:

**Search vsftpd**

Step6:

Command:

**use 1**

Step7:

Command:

**set RHOSTS 'ip' (eth0)**

Step8:

Command:

**run**

**Step9:**

**Command: search tomcat**